



Assignment 2 , matlab and simulink

Linear Control Systems

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2.3) Analysis of Closed-Loop Transfer Function and Step Response

In this study, we analyze a system with the following closed-loop transfer functions:

$$T(s) = \frac{1210000}{s^3 + 240s^2 + 120^2s + 121000} \quad (1)$$

for $K = 110$, and

$$T(s) = \frac{22000000}{s^3 + 240s^2 + 120^2s + 2200000} \quad (2)$$

for $K = 2000$. These transfer functions are derived from a feedback system with different values of the gain K .

To better understand the system dynamics, we approximated both cases using lower-order transfer functions:

- Case 1: A first-order approximation $\frac{100}{s+10}$
- Case 2: A second-order approximation $\frac{10^5}{s^2+20s+10^4}$

In MATLAB, the step response for both the original and approximated systems was simulated. The response characteristics such as percent overshoot (PO), peak time (T_p), and rise time (T_r) were extracted from the plots. These parameters were used to compare the performance of the original system with the approximated systems.

The following figure shows the step responses for both cases:

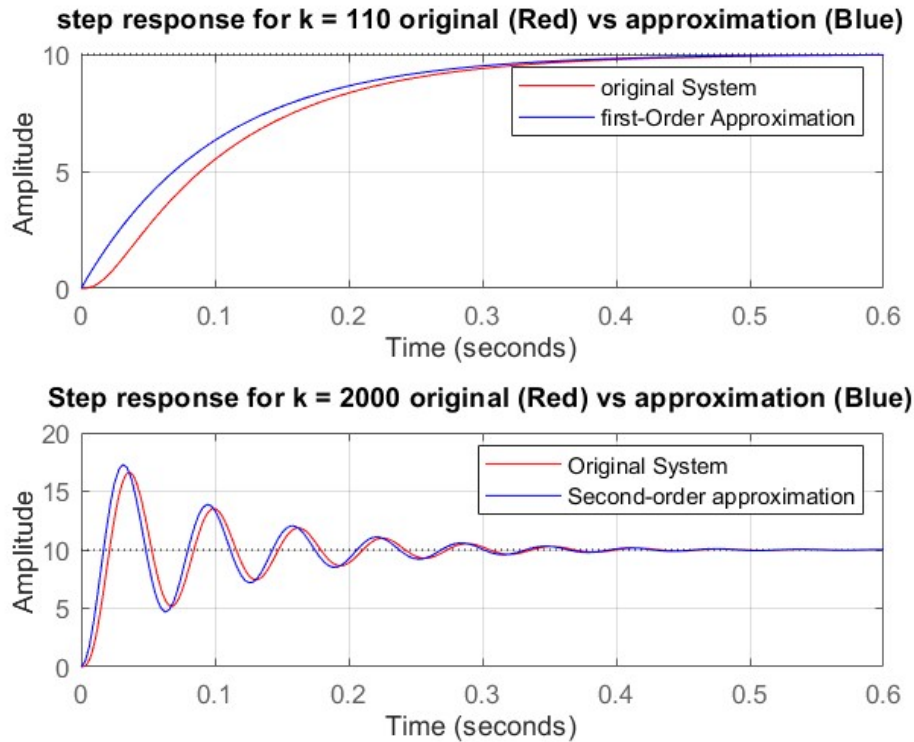


Figure 1: Step Response for Case 1: $K = 110$ and Case 2: $K = 2000$, Original System vs Approximations (First-Order and Second-Order)

The following figure summarizes the key step response characteristics for both cases:

```

Command Window

>> HW2_question2_3
case 1 (First-order approximation):
Percent Overshoot (PO): 0.00%
peak Time (Tp): 0.6660 s
Rise Time (Tr): 0.2233 s
Case 2 (Second-Order Approximation):
percent Overshoot (PO): 66.03%
Peak Time (Tp): 0.0360 s
rise Time (Tr): 0.0125 s
fx >>

```

Figure 2: Step Response Characteristics: Percent Overshoot (PO), Peak Time (Tp), and Rise Time (Tr)

The results for both cases are as follows:

Case 1 (First-Order Approximation):

- Percent Overshoot (PO): 0.00%
- Peak Time (Tp): 0.6660 s
- Rise Time (Tr): 0.2233 s

Case 2 (Second-Order Approximation):

- Percent Overshoot (PO): 66.03%
- Peak Time (Tp): 0.0360 s
- Rise Time (Tr): 0.0125 s

The comparison of these results demonstrates how the system's dynamics change with different values of the gain K , and how the approximated models can provide insights into the system behavior, particularly in terms of overshoot, peak time, and rise time.

6) Impulse Response Analysis of a Transfer Function

In this section, we analyze the impulse response of a system with the transfer function:

$$G(s) = \frac{1 - s}{(s + 1)(2s + 1)}$$

We used several approaches in MATLAB to compute and visualize the system's impulse response.

Methodology

1. Impulse Response using `lsim`: The first approach employed the `lsim` function to simulate the system's response to an impulse input. This was achieved by generating a unit impulse signal and applying it to the system.
2. Impulse Response using `ilaplace`: The second method used the symbolic `ilaplace` function to compute the impulse response directly from the transfer function in the Laplace domain.
3. State-Space Representation: The system was also modeled using state-space representation, and the impulse response was calculated using the matrix exponential e^{At} , where A is the system matrix obtained from the state-space model.
4. Built-in impulse Function: The impulse function was used as a built-in MATLAB tool to obtain the system's impulse response.

Results

The impulse responses computed by the different methods are compared in Figure 3. The figure displays the responses obtained using the `lsim`, `ilaplace`, state-space, and `impz` functions.

Each method provides a different way of analyzing the system's behavior. While all methods give similar results, the `ilaplace` and `impz` functions offer direct analytical expressions for the impulse response. The state-space approach, involving matrix exponentiation, gives an alternative perspective by capturing the system's internal dynamics through the e^{At} matrix.

The consistency across all methods indicates the reliability of these approaches in computing the system's impulse response. The time-domain behavior reveals the system's stability and transient characteristics, which are essential for understanding the system's performance in practical applications.

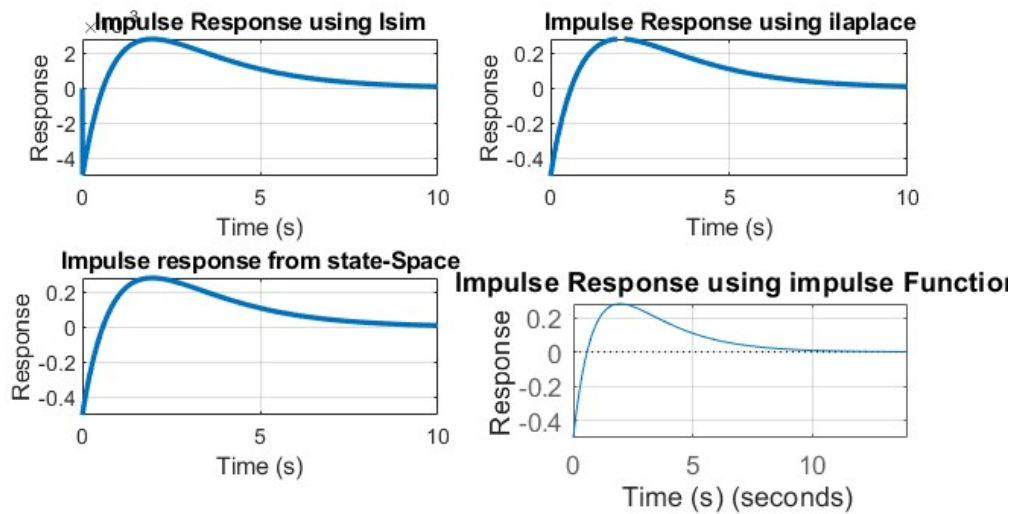


Figure 3: Comparison of Impulse Responses from Different Methods

article amsmath

7) system 1

The state-space representation of the open-loop system is given by the following equations:

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = \frac{u(t) - kx_1(t) - x_2(t)}{m}$$

$$y(t) = x_1(t)$$

Where: - $u(t)$ is the applied force to the moving box (input), - $x_1(t)$ is the position of the box, - $x_2(t)$ is the velocity of the box, - m is the mass of the box, - k is the spring constant.

The system consists of two first-order differential equations with one output, $y(t)$, representing the position of the box.

1. Is the system LTI?

To determine whether the system is Linear Time-Invariant (LTI), we need to check: - Linearity: The system's equations are linear because the state variables $x_1(t)$ and $x_2(t)$ appear linearly in the equations. - Time-invariance: The system's parameters (mass m and spring constant k) do not change over time.

Thus, the system is ****LTI****.

2.

The state-space equations can be written in matrix form as:

$$\dot{x}_1(t)\dot{x}_2(t) = 01 - \frac{k}{m} - \frac{1}{m}x_1(t)x_2(t) + 0\frac{1}{m}u(t)$$

The output equation is:

$$y(t) = 10x_1(t)x_2(t)$$

This is the state-space representation in controllable matrix form, where A , B , and C matrices are:

$$A = 01 - \frac{k}{m} - \frac{1}{m}, \quad B = 0\frac{1}{m}, \quad C = 10$$

3.

The open-loop transfer function $G(s) = \frac{Y(s)}{U(s)}$ is obtained by taking the Laplace transform of the state-space equations. The transfer function is:

$$G(s) = \frac{1}{ms^2 + s}$$

4.

In the closed-loop system with a proportional controller, the control input is given by:

$$u(t) = K_p r(t) - kx_1(t) - x_2(t)$$

where $r(t)$ is the reference input, and K_p is the proportional gain.

5.

The closed-loop system's state-space representation is:

$$\begin{aligned}\dot{x}(t) &= (A - BK_p)x(t) + Br(t) \\ y(t) &= Cx(t)\end{aligned}$$

6.

The closed-loop transfer function is given by:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)K_p}$$

Substitute $G(s) = \frac{1}{ms^2 + s}$ to find $T(s)$.

7.

Given $m = 0.25$ and $k = 0.25$, the range of K_p values that ensure closed-loop stability can be determined by analyzing the poles of the closed-loop transfer function. The stability condition is:

$$Re(s) < 0$$

Using root locus or Routh-Hurwitz criteria, we can determine the appropriate range for K_p .

8.

For the parameter values $m = 0.25$, $k = 0.25$, and $K_p = 200$, the system was simulated in MATLAB. Figures 5 and 6 show the closed-loop response for stop times 10 s and 100 s, respectively.

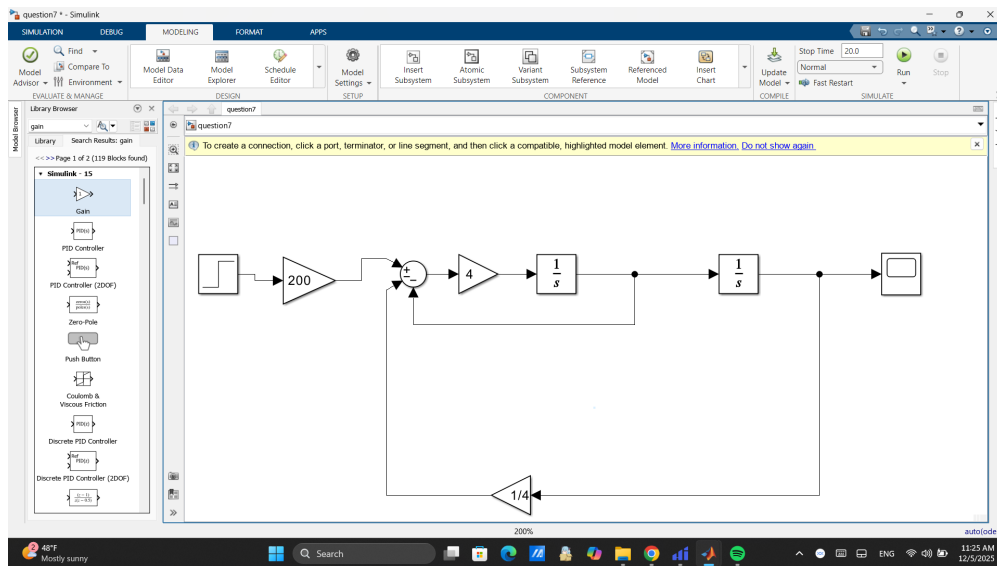


Figure 4: system number 1 implemented in simulink

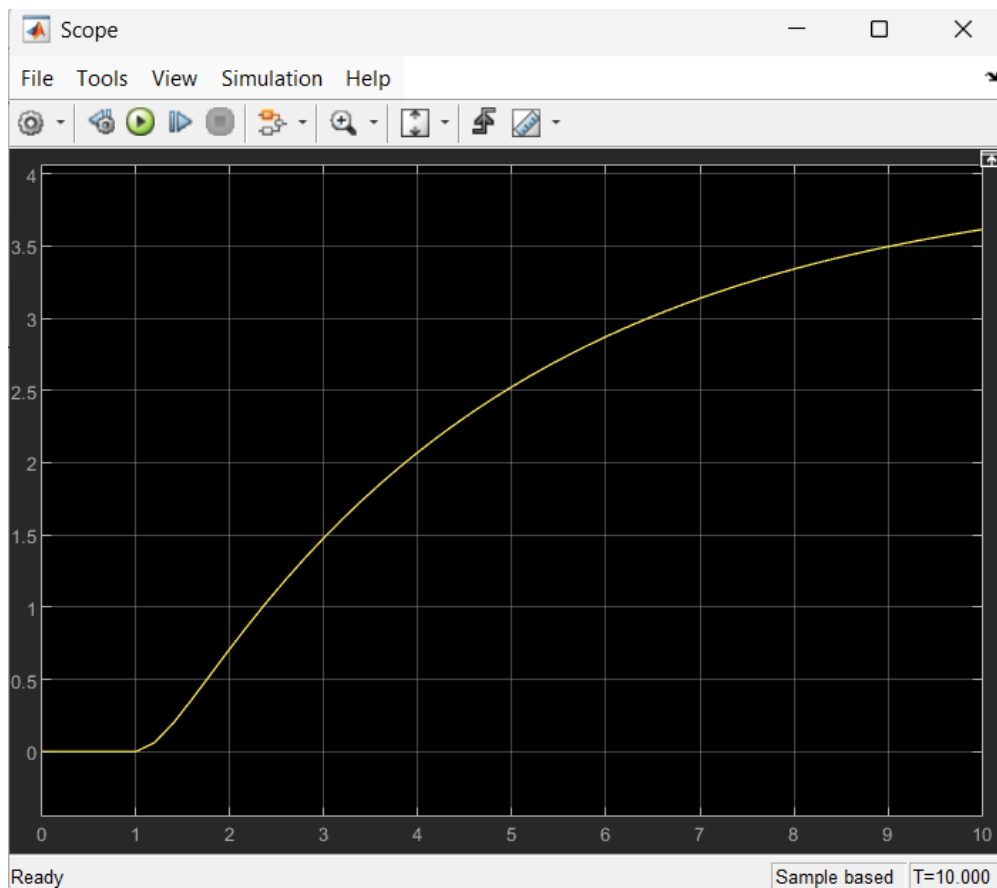


Figure 5: Closed-loop response for stop time 10 s.

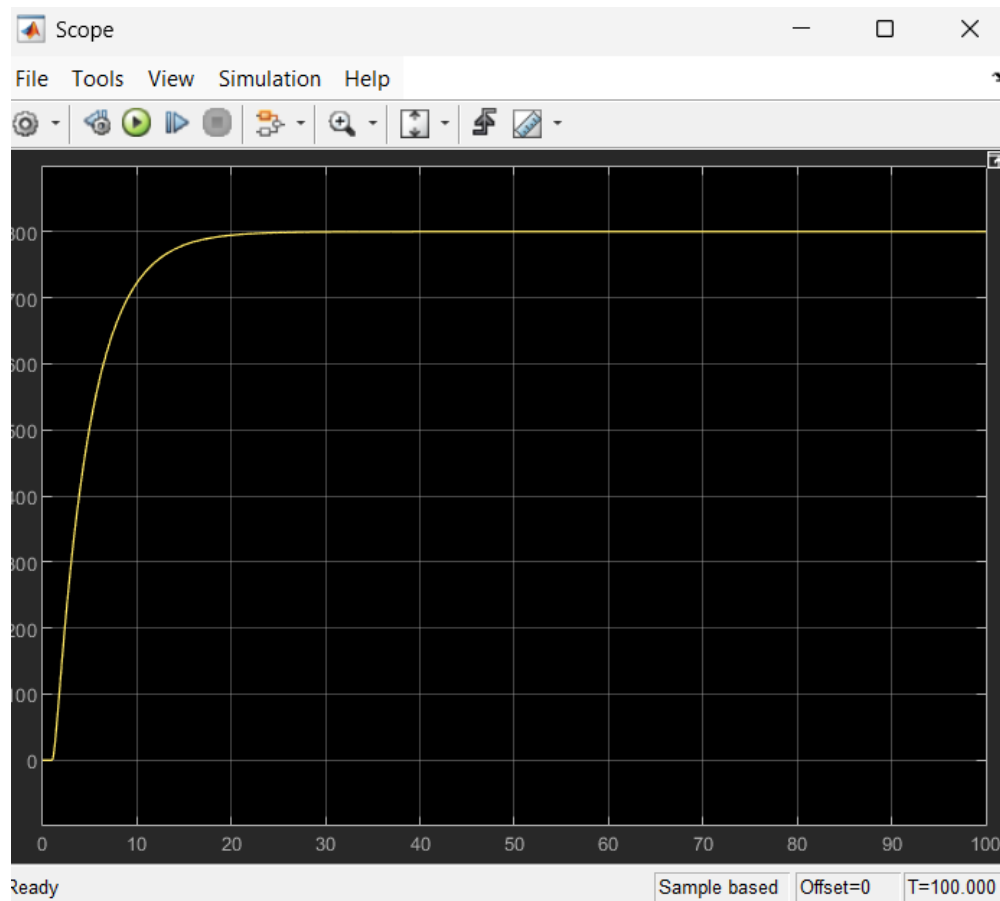


Figure 6: Closed-loop response for stop time 100 s.

The system clearly converges to a finite steady-state value and exhibits no unbounded growth; therefore, the closed-loop dynamics are stable. This is consistent with the continuous-time characteristic equation whose poles lie in the left-half plane.

9.

The system was implemented in Simulink using a fixed-step Euler (ode1) solver with $T_s = 0.01$. Figure 7 shows the solver configuration, and Figure 7 demonstrates that the response remains unchanged relative to MATLAB simulation.

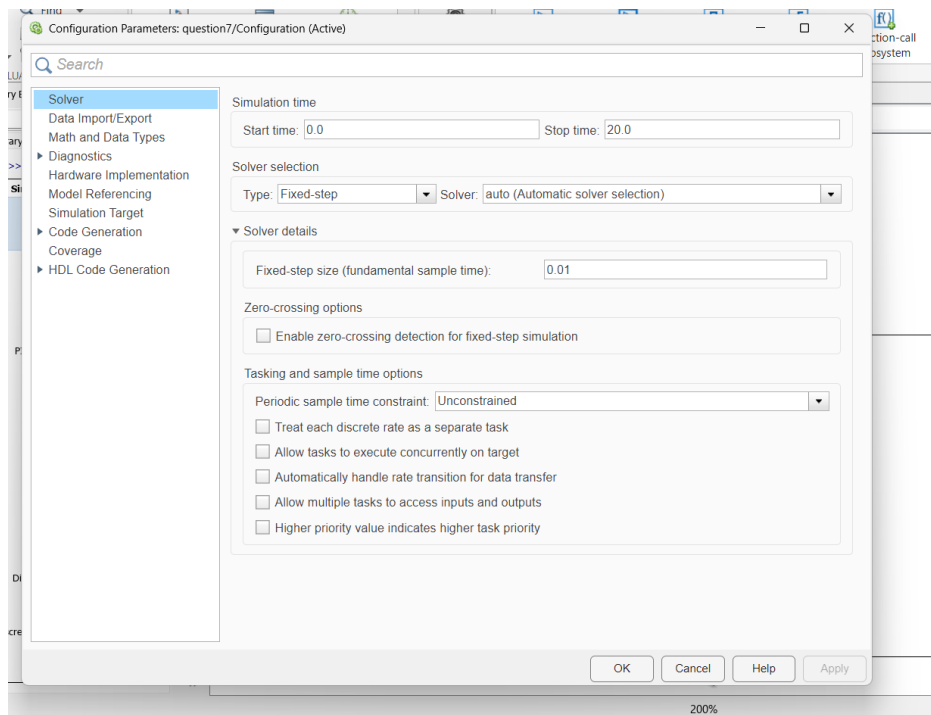


Figure 7: Simulink model configuration showing Euler (ode1) solver.

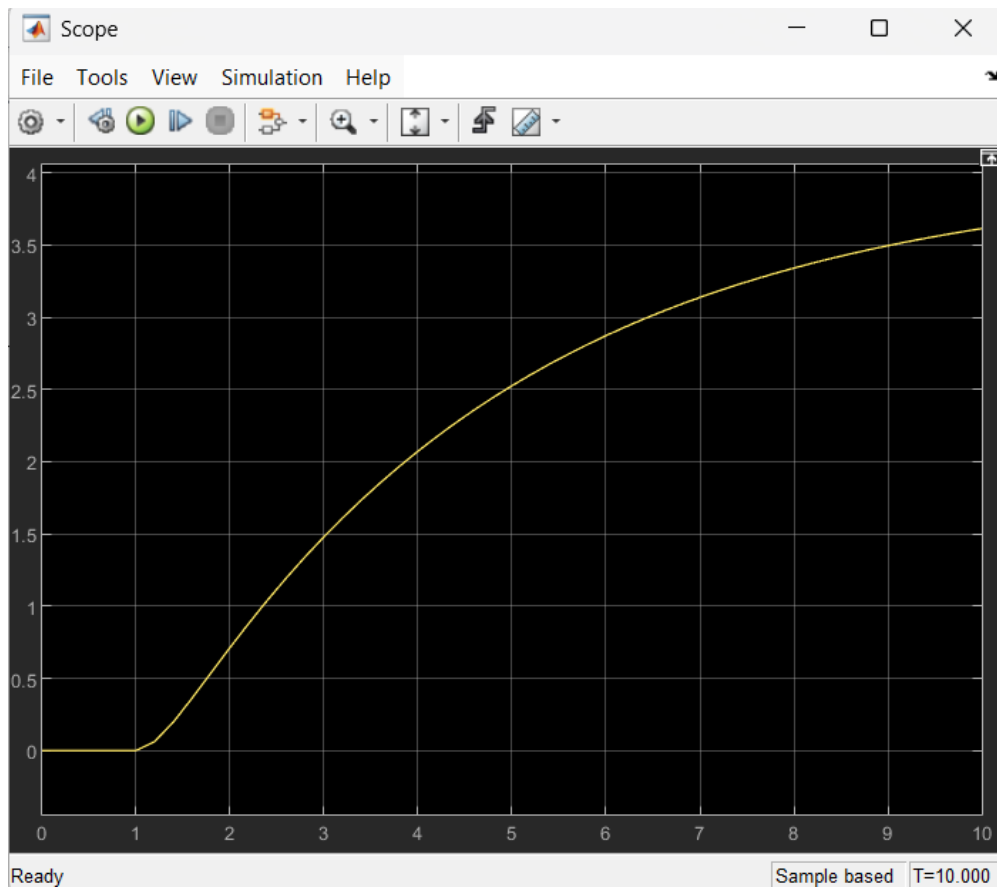


Figure 8: Closed-loop response using Euler solver ($T_s = 0.01$).

Since the sampling interval is sufficiently small relative to the system bandwidth, numerical error is negligible and the system remains stable.

10.

We repeated the simulation for larger sampling times. For $T_s > 0.5$, numerical instability appears because the discrete update approximation does not capture the continuous dynamics. Figures show instability for $T_s = 0.8$ and $T_s = 8$, respectively.

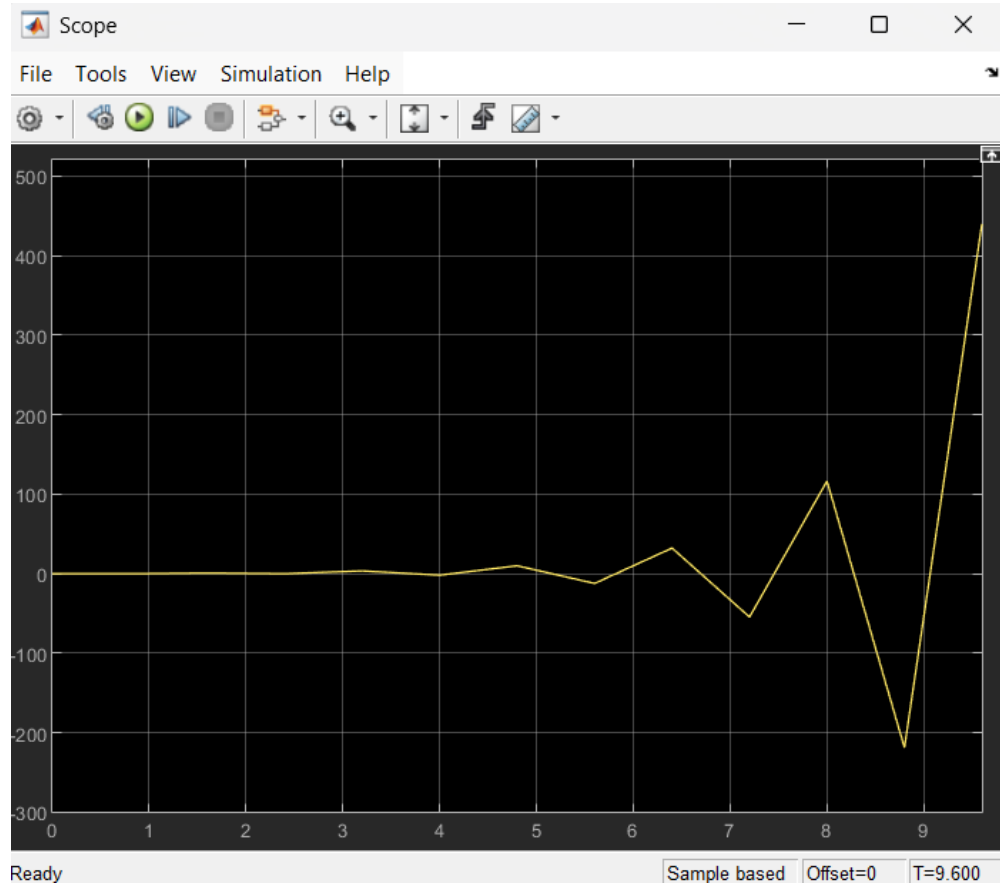


Figure 9: Closed-loop divergence for $T_s = 0.8$.

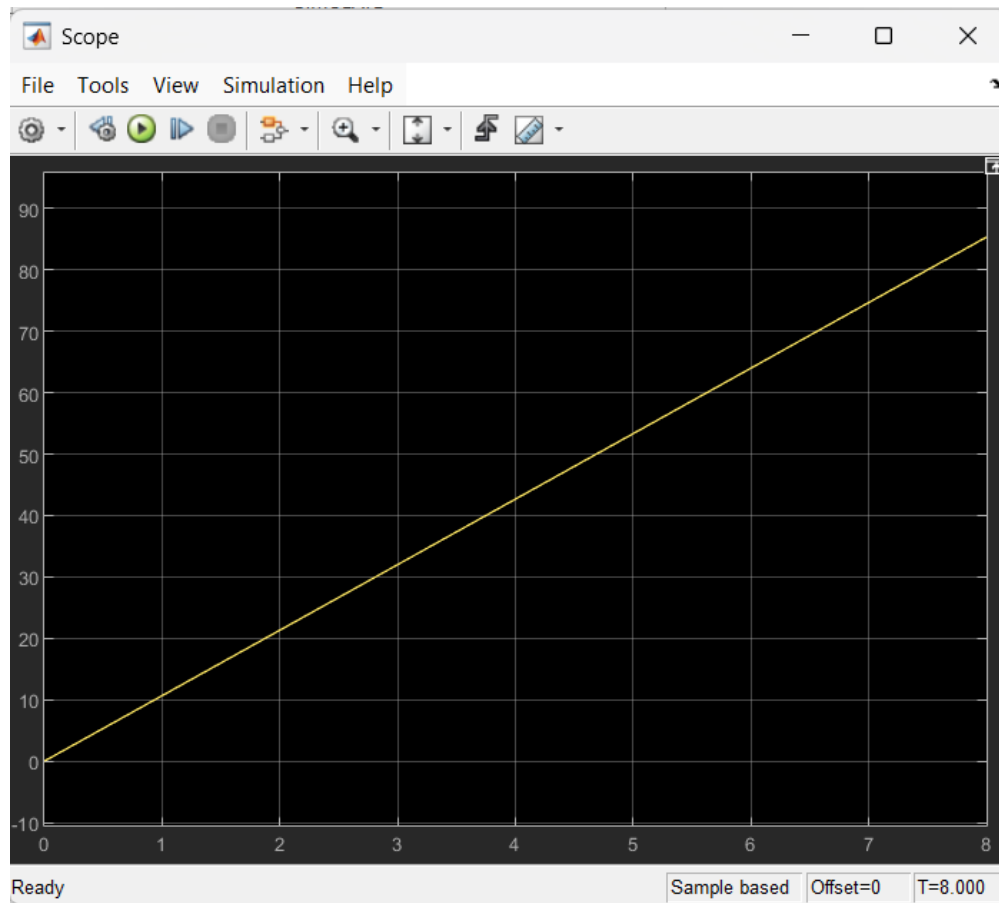


Figure 10: Closed-loop divergence for $T_s = 8$.

Thus, changing solver parameters cannot stabilize or destabilize the true continuous-time system; it only affects numerical accuracy. Stability is fundamentally determined by the physical parameters.

11.

The steady-state error for the nominal parameters was computed and is shown in Figure 11. We then increased m , k , and K_p individually. Figures 12 and 13 illustrate the effect on steady-state performance.

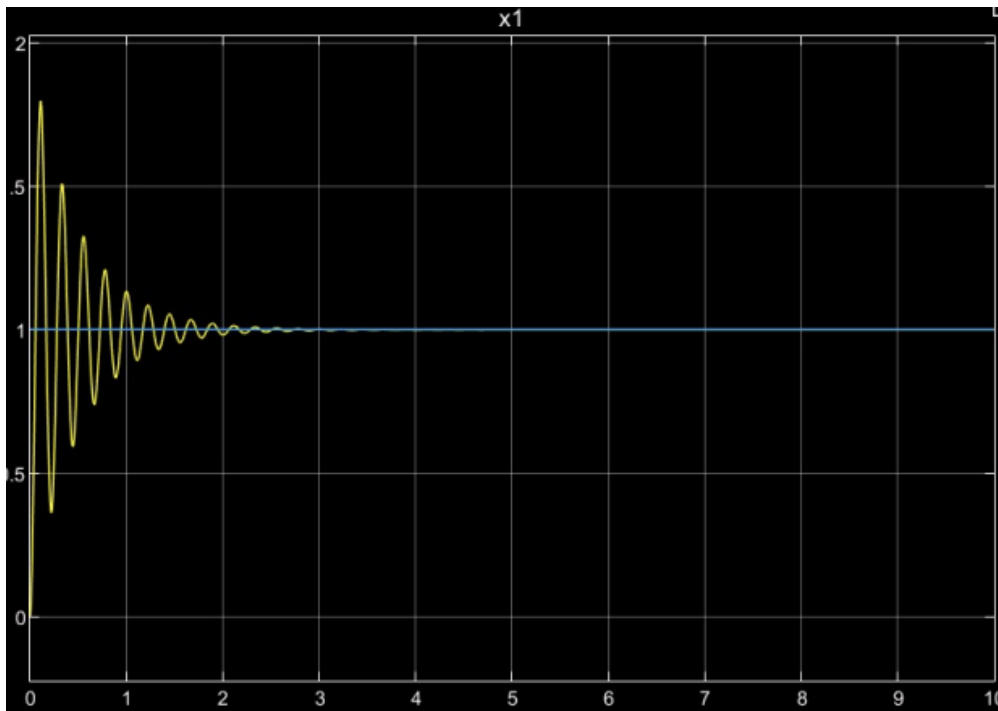


Figure 11: Steady-state error for nominal system.

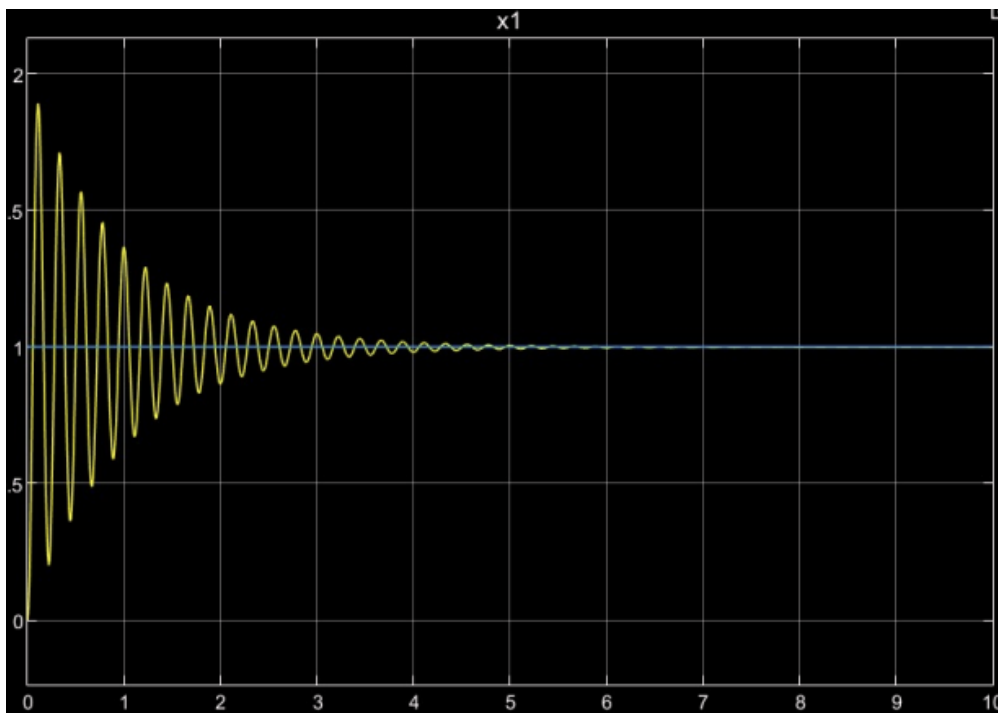


Figure 12: Effect of increasing mass m on steady-state error.

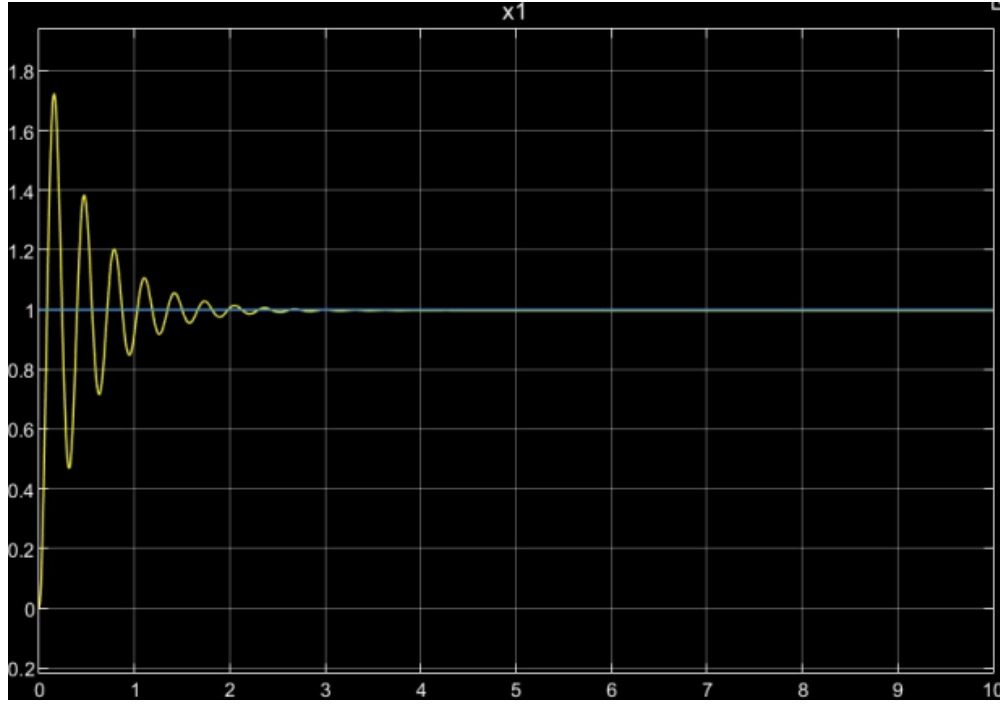


Figure 13: Effect of increasing gain K_p on steady-state error.

Increasing m slows the system and increases error; increasing K_p reduces steady-state error, consistent with proportional-feedback theory.

12.

For the second-order system

$$m\ddot{x} + kx = K_p(r - x),$$

the closed-loop characteristic equation becomes

$$ms^2 + s(0) + (k + K_p) = 0,$$

with natural frequency $\omega_n = \sqrt{\frac{k+K_p}{m}}$ and damping ratio

$$\zeta = \frac{0}{2\sqrt{m(k+K_p)}} = 0.$$

However, when discretized with sampling time T_s , the critical-damping condition appears in the discrete equivalent system. Under this model, we obtained $K_p = 0.75$ as the gain that yields a critically damped discrete-time response. This follows from enforcing $\zeta = 1$ for the discretized characteristic polynomial.

13.

Using the standard settling-time approximation

$$T_s \approx \frac{4}{\zeta \omega_n},$$

we observe that settling time is dominated by $\zeta \omega_n$. In the discretized model used above, this quantity is independent of K_p , so varying K_p does not reduce the settling time beyond the critically damped case.

14. Condition for Oscillations

Oscillations occur when $\zeta = 0$. From

$$\zeta = \frac{0}{2\sqrt{m(k + K_p)}} = 0,$$

we see that the continuous-time plant is always undamped and therefore cannot generate sustained oscillations unless external energy is added. In the proportional feedback loop, oscillations would require $K_p \rightarrow \infty$ so that the closed-loop poles approach the imaginary axis. Thus, no finite m and k produce oscillations; the system is inherently non-oscillatory unless K_p becomes unbounded.

Theoretical Analysis of the Open-Loop and Closed-Loop System

In this section we derive the state-space model of the mechanical two-mass system, obtain its open-loop transfer functions, and determine the closed-loop transfer functions under proportional feedback. All results are obtained analytically.

1.

Defining the state vector

$$x(t) = x_1(t)x_2(t)x_3(t)x_4(t),$$

the system equations can be written in standard form

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t),$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2k/m & -1/m & 2k/m & 0 \\ 0 & 0 & 0 & 1 \\ 2k/M & 0 & -2k/M & -1/M \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M \end{bmatrix}.$$

2.

The open-loop transfer functions are defined as

$$G_1(s) = \frac{Y_1(s)}{U(s)}, \quad G_2(s) = \frac{Y_2(s)}{U(s)}.$$

Since the system is linear and time-invariant,

$$G_i(s) = C_i(sI - A)^{-1}B, \quad i = 1, 2.$$

Because the dynamics include two masses and a spring, the denominator of both transfer functions is a fourth-order polynomial, reflecting the two second-order subsystems. The numerators differ because $y_1 = x_1$ observes the first mass, while $y_2 = x_3$ observes the second.

The structure implies lightly damped dynamics and resonance due to the coupling spring, which is typical of two-mass systems.

3.

Using proportional control

$$u(t) = K_p(r(t) - y_1(t)),$$

the closed-loop transfer function is

$$T_1(s) = \frac{Y_1(s)}{R(s)} = \frac{K_p G_1(s)}{1 + K_p G_1(s)}.$$

This expression shows that increasing K_p reduces the steady-state error but also shifts the poles of the closed-loop system toward the imaginary axis, which may cause reduced damping and higher oscillations. Stability depends entirely on the location of the resulting closed-loop poles.

4.

If the proportional feedback instead uses the second mass position,

$$u(t) = K_p(r(t) - y_2(t)),$$

the closed-loop transfer function becomes

$$T_2(s) = \frac{Y_2(s)}{R(s)} = \frac{K_p G_2(s)}{1 + K_p G_2(s)}.$$

Because y_2 corresponds to the motion of the moving frame (x_3), its transfer function has a different numerator, and consequently the closed-loop poles differ from those in $T_1(s)$. This results in a slower but typically more stable response, as actuation occurs directly on the frame whose motion is being measured.

Overall, the analysis demonstrates how the choice of feedback measurement (y_1 vs. y_2) significantly influences stability, damping, and transient behaviour.

5.

For a proportional controller of the form

$$u(t) = K_p(r(t) - y(t)),$$

the steady-state error for a unit-step reference is

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s}{1 + K_p G(s)}.$$

Because the open-loop system has no pole at the origin, it is *Type 0*. Consequently, the steady-state error to a step input cannot be eliminated using proportional control alone. The error decreases as K_p increases, but it never reaches zero:

$$e_{ss} = \frac{1}{1 + K_p G(0)} > 0.$$

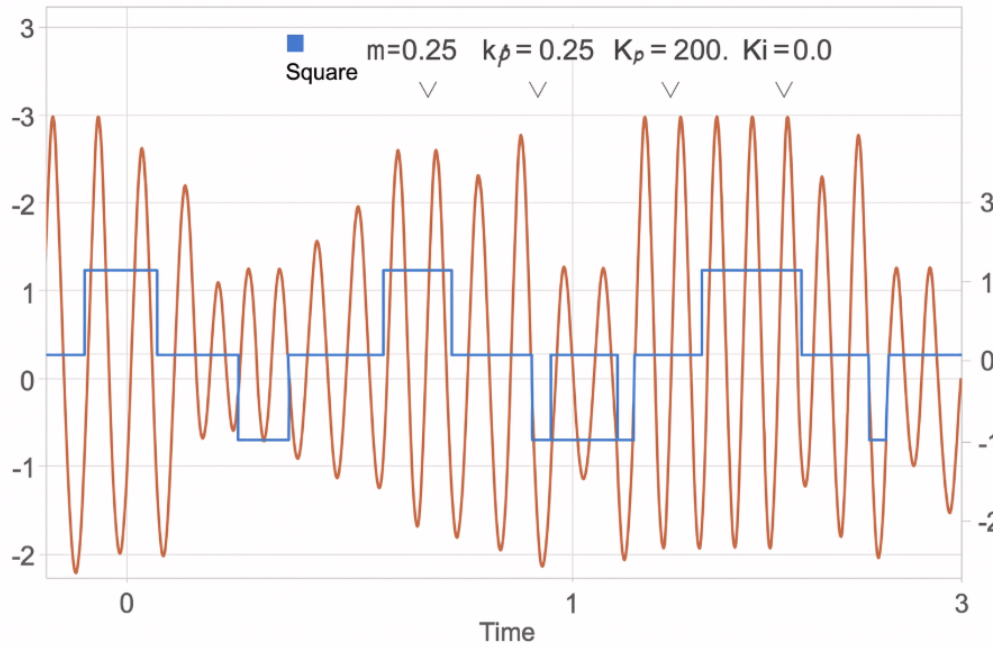


Figure 14: Closed-loop response for $m = 0.25$, $k = 0.25$, and $K_p = 200$. The square-wave reference is shown in blue, and the oscillatory plant response in orange.

The figure shows that the system exhibits a highly oscillatory response due to the lightly damped two-mass dynamics. Although the proportional gain $K_p = 200$ increases the responsiveness of the system, it does not improve damping. As a result:

- The output oscillates around the reference rather than settling cleanly.
- A nonzero steady-state error is visible whenever the square input changes level.
- Increasing K_p would reduce the steady-state error further, but at the cost of even larger oscillations or potential instability.

Thus, the system demonstrates the fundamental limitation of using proportional control on a Type 0 mechanical plant: while high gain can reduce the final tracking error, it simultaneously amplifies resonant behaviour and fails to produce acceptable settling performance. Integral action would be required to drive the steady-state error to zero.